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Data-driven evacuation and rescue traffic optimization with rescue contraflow control

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A B S T R A C T

In response to local sudden disasters, e.g., high-rise office or residential building fire disasters, road occupation can cause conflicts, and traffic directions may be opposite between evacuation vehicles and rescue vehicles; moreover, lane contraflow can be adopted to meet these surge traffic demands. However, lane contraflow that provides more roads for rescue vehicles reduces the traffic supply in the evacuation direction. It is unclear how to control the number of contraflow roads used by rescue vehicles to coordinate evacuation and rescue traffic operations. Here, we adjust the critical rescue traffic volume of reversing the normal road traffic direction to control rescue contraflow. Additionally, we propose a multiobjective mixed integer linear programming formulation for evacuation and rescue traffic optimization. Additionally, considering that the upper limit of the critical rescue traffic volume is unknown and that the proposed formulation includes multiple objectives and multi-priority vehicle classes, a three-stage solving algorithm is developed. Next, a large-scale evacuation and rescue traffic optimization result dataset is obtained for the Nguyen–Dupuis road network, and the impact of different rescue contraflow control plans on evacuation and rescue traffic is studied based on data-driven statistical analysis. The results show that by adjusting the optimal rescue traffic route, the critical rescue traffic volume for reversing the normal road traffic direction can reduce the interference of rescue traffic to evacuation traffic operation performance without reducing rescue traffic operation performance, and can be used to coordinate evacuation and rescue traffic operation under rescue contraflow.

1. Introduction

In real-world scenarios, disasters occurring in cities may be local or global. Local disasters, e.g., building fire disasters and dangerous chemical accidents, usually pose serious threats to people located only within the disaster position, and the corresponding disaster scope belongs to a very small part of the whole city, e.g., a building or a chemical plant. Global disasters, e.g., earthquakes, hurricanes, rainstorms, waterlogging, etc., usually impose serious threats on people located in whole cities. Moreover, some kinds of disasters may occur suddenly and unpredictably, e.g., building fire disasters, dangerous chemical accidents, earthquakes, etc., and other kinds of disasters can be forecasted in advance, e.g., hurricanes and rainstorm waterlogging. Unlike other disasters, sudden local disasters instantaneously cause surges to affect people and rescue resource demand in disaster positions, and surrounding roads are not damaged.

In response to local sudden disasters, e.g., high-rise office or residential building fire disasters, some affected people drive their private cars to leave the office or residential building from underground and ground parking, and rescue vehicles arrive at the disaster position to handle the disaster. In this situation, private evacuation vehicles for transferring affected people and emergency rescue vehicles for handling disasters are on the shared road network, and lane contraflow can be adopted to increase traffic supply and service surge traffic demand around the disaster position. However, evacuation vehicles and rescue vehicles travel in opposite traffic directions between the outside safe street area and the disaster position. They may simultaneously occupy the same roads and enter cross-conflict turns. Suppose some roads in the

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evacuation direction are reversed and allocated to rescue vehicles considering traffic priority under lane contraflow. In that case, the traffic supply to evacuation vehicles will decrease in the shared road network. Therefore, it is necessary to study evacuation and rescue traffic optimization considering rescue contraflow control, and provide support for collaborative planning of evacuation and rescue traffic operations.

1.1 Evacuation and rescue traffic optimization

In terms of evacuation and rescue traffic optimization in response to local sudden disasters, Chiu and Zheng [1] developed a single-objective linear programming model for dynamic traffic assignment of multi-priority vehicle groups to minimize the total prioritized travel time of evacuation and rescue traffic over the planning horizon. Xie and Turnquist [2] considered the travel time to determine the traffic route of emergency vehicles between the hospital and the accident site and then reserved the emergency traffic route during evacuation. They formulated the evacuation traffic network optimization problem as a bi-level mathematical programming model of discrete network design, and the underlying traffic flow pattern followed a stochastic user equilibrium to minimize the travel time of evacuation vehicles. Here, because the travel time of emergency vehicles does not reflect the impact of the dynamic traffic state, the traffic route of emergency vehicles was static, and the route of emergency vehicles did not change with the change in traffic state on roads. Yang et al. [3] adopted evacuation and rescue traffic priority parameters to weigh and minimize the number of evacuation and rescue vehicles abiding on different roads in the objective function of a single-objective mixed integer nonlinear programming model. Kimms and Maassen [4] developed a single-objective mixed integer linear programming model for minimizing the number of evacuation vehicles residing, leaving, or waiting on roads to optimize evacuation and rescue traffic plans on roads. Cui et al. [5] minimized the weighted values of evacuation cost, rescue cost, traffic conflict cost between evacuation vehicles and rescue vehicles, and link contraflow cost on the road network. They developed a single-objective mathematical programming model for optimizing evacuation and rescue traffic distribution patterns. Liu et al. [6] developed two single-objective linear programming models for optimizing evacuation and rescue traffic assignment plans under different rescue entrance openings. Liu et al. [7,8] studied multimode dynamic evacuation traffic planning considering buses and cars by developing a single-objective mathematical programming model. Rahman and Hasan [9] developed the dynamic graph convolutional long short-term memory neural network (DGCN-LSTM) model by collecting and analyzing large-scale traffic detector data during Hurricane Irma's evacuation for evacuation traffic prediction.

In these existing studies, evacuation and rescue traffic optimization can be summarized into three situations: (1) Obeying rescue traffic priority: Static rescue traffic routes are determined first and then reserved to eliminate road occupation conflicts during evacuation; subsequently, a single-objective mathematical programming formulation is developed to optimize evacuation traffic operation on the shared road network. Static rescue traffic routes do not reflect the effect of dynamic rescue traffic states, e.g., traffic congestion and speed, and are more suitable for low rescue traffic demand. (2) Weighting evacuation and rescue traffic travel costs (e.g., weighting the number of evacuation vehicles and rescue vehicles abiding on different roads) as a single-objective function does not guarantee rescue traffic priority in evacuation and rescue traffic networks, which may cause rescue vehicles to need to give way to evacuation vehicles in road occupation conflicts. In addition, weighting evacuation and rescue traffic travel costs does not reflect the independence between evacuation and rescue traffic optimization objectives. (3) Most researchers propose new optimization models and algorithms or focus on collecting, processing, and analyzing large-scale traffic observation and survey data under disaster scenarios. Moreover, only a few evacuation and rescue traffic scenarios are tested to validate the proposed optimization model and algorithm. Existing studies lack the use of optimization models and algorithms to obtain large datasets of evacuation and rescue traffic optimization results, and further investigation of general evacuation and rescue traffic operation collaborative mechanisms based on large-scale evacuation and rescue traffic optimization results is needed.

1.2 Evacuation and rescue traffic control

Road traffic control (e.g., signal control, cross-elimination, lane contraflow, etc.) affects traffic supply and efficiency. In terms of evacuation and rescue traffic control approaches, Xie et al. [10] discussed the dynamic evacuation network optimization problem involving the integration of lane contraflow and cross-elimination measures, and the corresponding optimization results revealed that implementing lane reversal on roadway sections and crossing elimination at intersections significantly decreased the total evacuation time and network clearance time. Zheng et al. [11] used traffic simulation to identify traffic bottlenecks and then adopted lane contraflow and signal control to eliminate these bottlenecks and improve evacuation performance. Hua et al. [12] studied the evacuation traffic network optimization problem involving the integration of lane contraflows and cross-elimination to improve traffic efficiency during large-scale evacuations. Yuan et al. [13] studied the best distribution of cross-elimination and signal control on evacuation traffic networks by developing a mathematical programming model. The results showed that the best distribution of uninterrupted flow and signalized intersections yields the minimum evacuation clearance time. Cui et al. [5] integrated lane contraflows to redesign road networks shared by evacuation vehicles and rescue vehicles, and ensure sufficient use of the traffic capacity of the road network during evacuation and rescue. Gong et al. [14] used the Nanjing "6.20″ road traffic accident as an example. Lane contraflow and signal control were taken comprehensively to carry out emergency traffic organization for the local urban road network. In two recent studies [15,16], the evacuation and rescue traffic optimization problem were also studied from the perspective of rescue contraflow and the nonunique nature of the optimal rescue traffic routes during the evacuation, and the impacts of both the number of contraflow roads occupied by rescue vehicles and multiple optimal rescue traffic routes sharing the same optimal objective function values on evacuation and rescue traffic were analyzed considering rescue traffic priority on the shared road network.

Overall, most of these existing studies assume that road traffic control approaches are applied to maximize the traffic operation performance of a specific class of vehicle among multi-class vehicles; additionally, they neglect some critical conditions for adopting corresponding road traffic control approaches that must be met and do not coordinate evacuation and rescue traffic operation on the shared road network. Taking lane contraflow as an example, for any two-way road consisting of road i and its opposite road m, when the rescue traffic demand reaches the preset critical rescue traffic volume on any road i, the traffic direction on its opposite road m can be reversed for rescue vehicles. Otherwise, the opposite road m is used in the normal road traffic direction. Obviously, rescue contraflow will provide more roads for rescue vehicles considering rescue priority, and the traffic supply along the evacuation direction will be reduced. The impact of rescue contraflow on evacuation and rescue traffic operations and how to control rescue contraflow to coordinate evacuation and rescue traffic operations are unclear.

1.3 Main motivation and contribution

Based on the gaps mentioned above in evacuation and rescue traffic control and optimization, we develop different rescue contraflow control plans by adjusting the critical rescue traffic volume for reversing normal road traffic directions and propose multiobjective evacuation and rescue traffic optimization formulations considering rescue contraflow control. Then, we design a three-stage algorithm to address the problems of unknown parameters, multiple objectives, and multipriority vehicle classes. Then, the impact of rescue contraflow on evacuation and rescue traffic is studied via data-driven statistical analysis based on the dataset collected via numerous evacuation and rescue traffic optimization steps. Finally, the novel conclusions drawn in this paper support collaborative planning for evacuation and rescue traffic operations of considering rescue contraflow during a sudden local disaster.

The main contributions of our work are as follows:

(1) We propose a multiobjective evacuation and rescue traffic optimization formulation of introducing rescue contraflow control by adopting different critical rescue traffic volumes of reversing the normal road traffic direction.
(2) We design a three-stage algorithm for solving the problem that evacuation and rescue traffic optimization formulation includes multiple objectives and multiple priority vehicle classes, and the upper limit values of some input parameters are unknown.
(3) We study the impact of rescue contraflow control on evacuation and rescue traffic by analyzing evacuation and rescue traffic optimization result datasets under different traffic demands and rescue traffic contraflow plans.
(4) We investigate the solutions for coordinating evacuation and rescue traffic operations via rescue contraflow control based on the dataset of large-scale evacuation and rescue traffic optimization results.

The remainder of this paper is organized as follows: Section 2 presents the proposed multiobjective evacuation and rescue traffic optimization formulation with rescue contraflow control. Section 3 presents the three-stage solving algorithm and the implementation process of obtaining the large-scale evacuation and rescue traffic optimization result dataset. Section 4 describes our numerical studies on the classical Nguyen–Dupuis road network based on data-driven statistical analysis. Finally, we present novel evacuation and rescue traffic optimization results with rescue contraflow control.

2. Evacuation and rescue traffic optimization formulation integrating rescue contraflow control

2.1 Mathematical notations

Here, evacuation and rescue traffic operation, road network structure, time window planning, and rescue contraflow control can be expressed mathematically with the following notation.

Notations	Descriptions
	Sets
Ω	Discrete simulation periods updating evacuation and rescue traffic states on the road network, $\Omega = \{1, 2, \ldots, T\}$
C	Multi-priorities vehicle classes consisting of evacuation vehicles and rescue vehicles, $C = \{0, 1\}$
N	Intersections constituting the road network around the disaster position
L	Roads constituting the road network around the disaster position
L_α^*	Roads used by rescue vehicles to enter the disaster position if $\alpha = 0$, or roads used by evacuation vehicles to leave the disaster position if $\alpha = 1$, $L_\alpha^* \subseteq L, \alpha \in C$
L_α°	Roads used by rescue vehicles to enter the surrounding road network of the disaster position from the outside safe street area if $\alpha = 0$, or roads used by evacuation vehicles to enter the outside safe street area from the surrounding road network of the disaster position if $\alpha = 1$, $L_\alpha^\circ \subseteq L, \alpha \in C$
L_R	Road occupied by rescue vehicles during the evacuation, $L_R \subseteq L$
Γ_i^-	Upstream adjacent roads of road i, $i \in L, \Gamma_i^- \subseteq L$
Γ_i^+	Downstream adjacent roads of road i, $i \in L, \Gamma_i^+ \subseteq L$
$\Gamma_n^{(i,j)}$	Turns conflicting with the turn from road i to road j at intersection n, $n \in N, i \in L, j \in \Gamma_i^+$
$\Gamma_{n,R}^{(i,j)}$	Evacuation traffic turns conflicting with rescue traffic turns from road i to road j at intersection n, $n \in N, i \in L, j \in \Gamma_i^+$
L_P	Paired roads of two-way roads, $L_P = \{\langle i,m \rangle \mid i,m \in L\}$, road i and road m constitute the same two-way road and have the same upstream and downstream intersections but opposite traffic direction
	Indices
t	Index of discrete simulation periods, $t \in \Omega$
i, j, m	Index of roads, $i, j, m \in L$
n	Index of intersections, $n = N$
α	Index of multi-priorities vehicle classes, $\alpha = 0$ stands for evacuation vehicles, and $\alpha = 1$ stands for rescue vehicles, $\alpha \in C$
	Parameters
T	Length of time window updating evacuation traffic state, $T = \lvert\Omega\rvert$
η	Length of any discrete simulation period, unit: second
l_i	Length of any road i, unit: meter, $i \in L$
τ_i	Travel time on road i under free-flow traffic states, unit: η, $i \in L$
ι_i	Propagation time of traffic congestion shockwave from downstream end to upstream end on road i, unit: η, $i \in L$
ρ_i^{jam}	Jam traffic density on road i, unit: vehicles/meter, $i \in L$
Q_i	Maximum number of the vehicles that can pass road i within any period t, unit: vehicles/η, $t \in \Omega, i \in L$
R	The number of rescue vehicles called from the outside safe street area
E	The number of private cars evacuating the affected people from the disaster position
	Variables
y_i	0–1 variable deciding rescue traffic direction on road i, $y_i = 0$ represents traffic direction on the road i does not be reversed for rescue vehicles; $y_i = 1$ represents traffic direction on the road i is reversed for rescue vehicles, $i \in L, y_i \in \{0, 1\}$
$y_{(i,j)}^{n,\alpha}$	0–1 variable deciding traffic turn from road i to road j at intersection n for evacuation vehicles and rescue vehicles, $y_{(i,j)}^{n,\alpha} = 0$ represents turn from road i to road j at intersection n is prohibited, and $y_{(i,j)}^{n,\alpha} = 1$ represents turn from road i to road j at intersection n is allowed, $n \in N$, $i \in L, j \in \Gamma_i^+, \alpha \in C$
t_i^-, t_i^+	The beginning and end period of that road i is occupied by rescue vehicles during the evacuation, $i \in L_R$
$t_{(i,j)}^{n,-}$, $t_{(i,j)}^{n,+}$	The beginning and end period that turn from road i to road j at intersection n is occupied by rescue vehicles during the evacuation, $n \in N, i, j \in L_R, j \in \Gamma_i^+$
$x_{i,\alpha}^t$	The number of evacuation vehicles and rescue vehicles on the road i at the beginning of period t, unit: vehicles, $t \in \Omega, i \in L, \alpha \in C$
$f_{(i,j)}^{t,\alpha}$	The number of evacuation vehicles and rescue vehicles entering road$j$from road i within period t, unit: vehicles, $t \in \Omega, i \in L, j \in \Gamma_i^+, \alpha \in C$
$U_{i,\alpha}^t$	The cumulative number of evacuation vehicles and rescue vehicles entering road i by the end of period t, unit: vehicles, $t \in \Omega, i \in L, \alpha \in C$
$V_{i,\alpha}^t$	The cumulative number of evacuation vehicles and rescue vehicles leaving road i by the end of period t, unit: vehicles, $t \in \Omega, i \in L, \alpha \in C$
$G_{i,0}^t$	The number of rescue vehicles entering the disaster position from road i within period t, unit: vehicles, $t \in \Omega, i \in L_0^*$
$G_{i,1}^t$	The number of evacuation vehicles leaving the disaster position from road i within period t, unit: vehicles, $t \in \Omega, i \in L_1^*$
$E_{i,0}^t$	The number of rescue vehicles entering road i from the outside safe street area within period t, unit: vehicles, $t \in \Omega, i \in L_0^\circ$
$E_{i,1}^t$	The number of evacuation vehicles arriving at the outside safe street area from road i within period t, unit: vehicles, $t \in \Omega, i \in L_1^\circ$

2.2 LTM-based evacuation and rescue traffic simulation formulation

Dynamic network traffic flow is formed by the operation of vehicles on adjacent roads and the updating of traffic states (e.g., traffic density and traffic speed) over time; this process can be simulated by the link transmission model (LTM) proposed by Yperman et al. [17,18]. In the LTM, the number of vehicles ($f_{(i,j)}^{t,\alpha}$) entering road j from upstream adjacent road i within any period t can be determined by the sending traffic flow ($S_{i,\alpha}^t$) on road i and the receiving traffic flow ($R_{i,\alpha}^t$) on road j (see Eq. (1)) based on Newell's simplified theory [19–21].

$$f_{(i,j)}^{t,\alpha} = \min\left\{S_{i,\alpha}^t, R_{j,\alpha}^t\right\} \qquad t \in \Omega, i \in L, j \in \Gamma_i^+, \alpha \in C \tag{1}$$

In Eq. (1), within any period t, the sending traffic flow indicates that the vehicle can leave road i, in which the corresponding traffic volume can be determined by the road capacity Q_i and the free-flow traffic state on road i (see Eq. (2)); the receiving traffic flow indicates that the vehicle can enter road j, in which the corresponding traffic volume can be determined by the road capacity Q_j and the remaining space not occupied by the vehicle on road j (see Eq. (3)). Here if we relax the LTM defined by Eqs. (1-4) can be obtained.

$$S^t_{i,\alpha} = \min\left\{U^{t-\tau_i}_{i,\alpha} - V^{t-1}_{i,\alpha}, Q_i\right\} \qquad t\in\Omega, i\in L, \alpha\in C \tag{2}$$

$$R^t_{j,\alpha} = \min\left\{V^{t-\iota_j}_{j,\alpha} + l_j\rho^{jam}_j - U^{t-1}_{j,\alpha}, Q_j\right\} \qquad t\in\Omega, j\in L, \alpha\in C \tag{3}$$

$$f^{t,\alpha}_{(i,j)} \le \begin{cases} U^{t-\tau_i}_{i,\alpha} - V^{t-1}_{i,\alpha} \\ Q_i \\ V^{t-\iota_j}_{j,\alpha} + l_j\rho^{jam}_j - U^{t-1}_{j,\alpha} \\ Q_j \end{cases} \xrightarrow{f^{t,\alpha}_{(i,j)} = V^t_{i,\alpha} - V^{t-1}_{i,\alpha}} \begin{cases} V^t_{i,\alpha} \le U^{t-\tau_i}_{i,\alpha} \\ V^t_{i,\alpha} - V^{t-1}_{i,\alpha} \le Q_i \\ V^t_{i,\alpha} \le V^{t-\iota_j}_{j,\alpha} + l_j\rho^{jam}_j \\ V^t_{i,\alpha} - V^{t-1}_{i,\alpha} \le Q_j \end{cases} \quad t \in\Omega, i\in L, j\in\Gamma^+_i, \alpha\in C \tag{4}$$

2.3 Cross-conflict turn elimination and lane contraflow formulation

Cross-conflict turns exist at intersections, which can be eliminated by Eqs. (5) and (6). In Eq. (6), $M \ge \min\{Q_i, Q_j\}$.

$$y^{n,\alpha}_{(i,j)} + y^{n,\alpha}_{(k,l)} \le 1 \qquad n\in N, i,k\in L, j\in\Gamma^+_i, l\in\Gamma^+_k, (k,l)\in\Gamma^{(i,j)}_n \tag{5}$$

$$f^{t,\alpha}_{(i,j)} \le M y^{n,\alpha}_{(i,j)} \qquad t\in\Omega, n\in N, i\in L, j\in\Gamma^+_i, \alpha\in C \tag{6}$$

In practice, lane contraflow can increase the traffic supply in special traffic directions. If the traffic direction on the road i is reversed ($y_i = 1$), the traffic direction on its paired road m should not be reversed ($y_m = 0$) (see Eq. (7)).

$$y_i + y_m \le 1 \qquad i, m\in L, \langle i,m\rangle \in L_P \tag{7}$$

2.4 Multiobjective mixed integer linear programming (MMILP) formulation for evacuation and rescue traffic optimization and control

In response to local sudden disasters, e.g., high-rise offices or residential building fire disasters, rescue vehicles handling disasters should quickly arrive at disaster positions from outside safe street areas, and evacuation vehicles transferring affected people should also quickly arrive at outside safe street areas from disaster positions. Here, these objectives can be expressed mathematically as Eq. (8) and Eq. (9) to evaluate the rescue and evacuation traffic operation performance.

$$\max\ Z_0(E,R,\lambda) = \sum_{t=1}^{T}\sum_{\tau=1}^{t}\sum_{i\in L^*_0} G^\tau_{i,0} \tag{8}$$

$$\min\ Z_1(E,R,\lambda) = \sum_{t=1}^{T}\left(E - \sum_{\tau=1}^{t} G^\tau_{i,1}\right) + \sum_{t=1}^{T+1}\sum_{i\in L} x^t_i \tag{9}$$

Here, $Z_0(E,R,\lambda)$ is used to maximize the objective function Z_0 of rescue traffic optimization, and $Z_1(E,R,\lambda)$ is used to minimize the objective function Z_1 of evacuation traffic optimization when the value of evacuation traffic demand is E, the value of rescue traffic demand is R, and the value of the rescue contraflow control parameter is λ (see Eq. (34)). Without loss of generality if we assume that t_1 and t_2 ($t_1 < t_2, t_1, t_2 \in \Omega$) are two feasible periods for all G rescue vehicles, the corresponding objective function values are $\sum_{t=t_1}^{T} G$ and $\sum_{t=t_2}^{T} G$, respectively, in Eq. (8). Obviously, $\sum_{t=t_1}^{T} G = \sum_{t=t_1}^{t_2-1} G + \sum_{t=t_2}^{T} G$ is greater than $\sum_{t=t_2}^{T} G$; thus, Eq. (8) can guarantee that rescue vehicles arrive at the disaster position as quickly as possible by obtaining the optimal objective function value. In Eq. (9), minimizing the number of evacuation vehicles stranded both in the disaster area and on the road network at the end of any period t guarantees that the affected people are quickly transferred to the outside safe street area from the disaster position. In addition, to solve Eqs. (8) and (9), evacuation and rescue traffic optimization and control considering rescue contraflow and rescue priority should also meet the following constraints.

2.4.1 Evacuation and rescue traffic conservation constraints

The traffic conservation constraints update the traffic state according to the number of evacuation vehicles and rescue vehicles on roads with the change in period, which can be expressed as Eq. (10). If $\alpha = 0$, Eq. (10) represents the number of rescue vehicles on the road i at the beginning of period $t+1$ is equal to the number of rescue vehicles on the road i at the beginning of period t plus the number of rescue vehicles entering road i from its upstream adjacent roads and the outside safe street area, and minus the number of rescue vehicles entering the downstream adjacent roads and the disaster position. If $\alpha = 1$, Eq. (10) represents the number of evacuation vehicles on road i at the beginning of period $t+1$ is equal to the number of evacuation vehicles on road i at the beginning of period t plus the number of evacuation vehicles entering road i from its upstream adjacent roads and the disaster position, and minus the number of evacuation vehicles entering the downstream adjacent roads and the outside safe street area.

$$x^{t+1}_{i,\alpha} = x^t_{i,\alpha} + \left(\sum_{j\in\Gamma^-_i} f^{t,\alpha}_{(j,i)} + (2\alpha-1)G^t_{i,\alpha}) - (\sum_{j\in\Gamma^+_i} f^{t,\alpha}_{(i,j)} + (2\alpha-1)E^t_{i,\alpha}\right) \qquad i \in L, \alpha\in C \tag{10}$$

2.4.2 Evacuation and rescue traffic dynamic loading constraints

Based on the relaxed LTM shown in Eq. (4), the traffic operation of evacuation vehicles and rescue vehicles on the road network can be modeled and calculated mathematically by Eqs. (11)–(14).

$$V^t_{i,\alpha} \le U^{t-\tau_i}_{i,\alpha} \qquad t\in\{\lfloor\tau_i\rfloor+1, \lfloor\tau_i\rfloor+2, \cdots, T\}, i\in L, \alpha\in C \tag{11}$$

$$V^t_{i,\alpha} - V^{t-1}_{i,\alpha} \le Q_i \qquad t\in\Omega, i\in L, \alpha\in C \tag{12}$$

$$U^t_{i,\alpha} \le V^{t-\iota_i}_{i,\alpha} + l_i\rho^{jam}_i \qquad t\in\{\lfloor\iota_i\rfloor+1, \lfloor\iota_i\rfloor+2, \cdots, T\}, i\in L, \alpha\in C \tag{13}$$

$$U^t_{i,\alpha} - U^{t-1}_{i,\alpha} \le Q_i \qquad t\in\Omega, i\in L, \alpha\in C \tag{14}$$

Based on Eqs. (11)-(14), when optimizing the dynamic transfer process of evacuation vehicles and rescue vehicles on the road network, the traffic volume among different roads within any period t should meet the constraints shown by Eqs. (15) and (16). Eq. (15) obtains the number of rescue vehicles and evacuation vehicles entering any road i within any period t, and Eq. (16) obtains the number of rescue vehicles and evacuation vehicles leaving any road i within any period t.

$$U^t_{i,\alpha} - U^{t-1}_{i,\alpha} = \sum_{j\in\Gamma^-_i} f^{t,\alpha}_{(j,i)} + (1-\alpha)E^t_{i,\alpha} + \alpha G^t_{i,\alpha} \qquad t\in\Omega, i\in L, \alpha\in C \tag{15}$$

$$V^t_{i,\alpha} - V^{t-1}_{i,\alpha} = \sum_{j\in\Gamma^+_i} f^{t,\alpha}_{(i,j)} + (1-\alpha)G^t_{i,\alpha} + \alpha E^t_{i,\alpha} \qquad t\in\Omega, i\in L, \alpha\in C \tag{16}$$

In Eqs. (15)and (16), period t is defined as the integer. Here, linear interpolation is used to calculate the cumulative number of evacuation and rescue vehicles entering and leaving any road i by any period $t+\delta(\delta\in[0,1])$ (see Eqs. (17) and (18)).

$$U^{t+\delta}_{i,\alpha} = (1-\delta)U^t_{i,\alpha} + \delta U^{t+1}_{i,\alpha} \qquad t\in\Omega, i\in L, \alpha\in C \tag{17}$$

$$V^{t+\delta}_{i,\alpha} = \begin{cases} (1-\delta)V^t_{i,\alpha} & t = \lfloor\tau_i\rfloor, 0<\delta<\tau_i - \lfloor\tau_i\rfloor, i\in L, \alpha\in C \\ \dfrac{\delta-(\tau_i - t)}{1-(\tau_i - t)} V^t_{i,\alpha} & t = \lfloor\tau_i\rfloor, \tau_i - \lfloor\tau_i\rfloor < \delta \le 1, i\in L, \alpha\in C \\ (1-\delta)V^t_{i,\alpha} + \delta V^t_{i+1,\alpha} & t \ne \lfloor\tau_i\rfloor, i\in L, \alpha\in C \end{cases} \tag{18}$$

Here, when loading evacuation and rescue traffic among different roads, evacuation vehicles need to enter the road network from the disaster position and finally arrive at the outside safe street area; rescue vehicles need to enter the road network from the outside and finally arrive at the disaster position. Eqs. (19) and (20) present the mathematical expression of rescue and evacuation traffic generation and the attraction balance process.

$$(1-\alpha)R+\alpha E=(1-\alpha)\sum_{t=1}^{T}\sum_{i\in L_\alpha^*}E_{i,\alpha}^t+\alpha\sum_{t=1}^{T}\sum_{i\in L_\alpha^*}G_{i,\alpha}^t \qquad L_\alpha^*, L_\alpha^* \subseteq L, \alpha\in C \tag{19}$$

$$(1-\alpha)R+\alpha E=\alpha\sum_{t=1}^{T}\sum_{i\in L_\alpha^*}E_{i,\alpha}^t+(1-\alpha)\sum_{t=1}^{T}\sum_{i\in L_\alpha^*}G_{i,\alpha}^t \qquad L_\alpha^*, L_\alpha^* \subseteq L, \alpha\in C \tag{20}$$

In addition, the decision variable of simulating evacuation and rescue traffic operations among different roads should meet the following nonnegative constraints. Before evacuation and rescue work, there is no evacuation or rescue traffic on the road network (see Eqs. (21) and (22)). Then, constrained by free-flow travel time, the vehicles do not pass their current roads (e.g., road i) within τ_i periods (see Eq. (23)). During evacuation and rescue, the number of vehicles traveling along adjacent roads should not be less than 0 (please refer to Eqs. (24)–(27)).

$$U_{i,\alpha}^0=0 \qquad i\in L,\alpha\in C \tag{21}$$

$$x_{i,\alpha}^1=0 \qquad i\in L,\alpha\in C \tag{22}$$

$$V_{i,\alpha}^t=0 \qquad t\in\{1,2,\ldots,\lfloor\tau_i\rfloor\}, i\in L,\alpha\in C \tag{23}$$

$$G_{i,\alpha}^t\geq 0 \qquad t\in\Omega, i\in L,\alpha\in C \tag{24}$$

$$E_{i,\alpha}^t\geq 0 \qquad t\in\Omega, i\in L,\alpha\in C \tag{25}$$

$$f_{(i,j)}^{t,\alpha}\geq 0 \qquad t\in\Omega, i\in L, j\in\Gamma_i^+,\alpha\in C \tag{26}$$

$$x_{i,\alpha}^t\geq 0 \qquad t\in\Omega, i\in L,\alpha\in C \tag{27}$$

2.4.3 Rescue traffic priority constraints

Considering road occupation conflicts, rescue vehicles usually have traffic priority, and evacuation vehicles need to wait for the passage of rescue vehicles. Eqs. (28) and (29) are rescue traffic priority constraints. Eq. (28) represents that evacuation vehicles are prohibited from entering the roads occupied by rescue vehicles until rescue vehicles pass them, and Eq. (29) represents that if evacuation traffic conflicts with rescue traffic turns at intersections, evacuation vehicles are prohibited from entering the conflict turns until rescue vehicles pass the intersections.

$$U_{i,1}^{t_i^+}-U_{i,1}^{t_i^- -1}=0 \qquad t_i^+,t_i^-\in\Omega, i\in L_R \tag{28}$$

$$\sum_{t=t_{(i,j)}^{n,-}}^{t_{(i,j)}^{n,+}}\sum_{(k,l)\in\Gamma_{n,R}^{(i,j)}}f_{(k,l)}^{t,1}=0 \qquad n\in N, i,j\in L_R, j\in\Gamma_i^+ \tag{29}$$

2.4.4 Rescue contraflow control constraints

Eqs. (30)–(33) represent LTM-based rescue traffic operation constraints under lane contraflow.

$$V_{i,0}^t\leq U_{i,0}^{t-\tau_i} \qquad t\in\{\lfloor\tau_i\rfloor+1,\lfloor\tau_i\rfloor+2,\cdots,T\}, i\in L \tag{30}$$

$$V_{i,0}^t-V_{i,0}^{t-1}\leq(1-y_i)Q_i+y_mQ_m \qquad t\in\Omega, i,m\in L,\langle i,m\rangle\in L_P \tag{31}$$

$$U_{i,0}^t\leq V_{i,0}^{t-\iota_i}+(1-y_i)l_i\rho_i^{jam}+y_ml_m\rho_m^{jam} \qquad t\in\{\lfloor\iota_i\rfloor+1,\lfloor\iota_i\rfloor+2,\cdots,T\},\langle i,m\rangle\in L_P \tag{32}$$

$$U_{i,0}^t-U_{i,0}^{t-1}\leq(1-y_i)Q_i+y_mQ_m \qquad t\in\Omega, i,m\in L,\langle i,m\rangle\in L_P \tag{33}$$

Obviously, $y_i=0, y_m=1$ but $U_{i,0}^t-U_{i,0}^{t-1}\leq Q_i$ is one of the feasible solutions in Eqs. (7) and (33), which means that if the feasible solution is used as the optimal rescue contraflow plan on evacuation and rescue traffic networks considering rescue priority, one unreasonable situation that occupies traffic supply along the evacuation direction will occur: the number of rescue vehicles entering road i is not more than its road capacity in the normal road traffic direction. Still, its paired road m is reversed in the traffic direction. Moreover, if lane contraflows are used by rescue vehicles, the traffic supply in the evacuation direction will decrease, and rescue contraflows may even delay evacuation traffic.

Here, by introducing parameter λ, we adopt the critical rescue traffic volume of reversing the normal road traffic direction to control rescue contraflow and then develop Eq. (34) to guarantee that if the paired road m of any road i need to be reversed in the traffic direction, the corresponding rescue traffic demand of using road i should be not less than the critical rescue traffic volume of reversing the normal road traffic direction (see Eq. (34)). In Eq. (34), λy_mQ_i is the critical rescue traffic volume for reversing the traffic direction on road m, and λ is the rescue contraflow control parameter. Here, λ is not less than 0, and changing the rescue contraflow control parameter can adjust the critical rescue traffic volume of reversing the normal road traffic direction.

$$U_{i,0}^T\geq\lambda y_mQ_i \qquad i,m\in L,\langle i,m\rangle\in L_P \tag{34}$$

Eq. (35) is added to prohibit evacuation vehicles from entering contraflow roads occupied by rescue vehicles.

$$U_{m,1}^{t_i^+}-U_{m,1}^{t_i^-}=0 \qquad t_i^+,t_i^-\in\Omega, i\in L_R,\langle i,m\rangle\in L_P \tag{35}$$

3. Three-stage solving algorithm

In Eq. (34), the minimum value of λ is 0, which indicates that there is no rescue traffic demand restriction for reversing the normal road traffic direction; however, the maximum value of λ (in which the rescue traffic operation performance is equal to the performance of not taking lane contraflow into account) is not determined. The corresponding upper limit value of λy_mQ_i indicates that all roads are in the normal traffic direction and that there are no contraflow roads for rescue vehicles. Considering that the upper limit of the critical rescue traffic volume of reversing normal road traffic direction is unknown, solving the proposed MMILP formulation involves several difficulties:

(1) Evacuation and rescue vehicles operate on the shared road network within the same time window, and two independent optimization objectives exist between two different classes of vehicles. (2) On the one hand, the optimal evacuation traffic route and the optimal rescue traffic route should be simultaneously generated; on the other hand, because of road occupation conflicts and rescue traffic priorities, the optimal rescue traffic route needs to be known in advance and then reserved to eliminate traffic conflicts during evacuation. (3) For any rescue traffic demand, when increasing the critical rescue traffic volume of reversing the normal road traffic direction, we do not know whether and when we should stop using lane contraflows for rescue vehicles on roads to avoid a decrease in rescue traffic operation performance.

Table 1
Three-stage solving algorithm.

Stage 1:	max Z_0	s.t. Eqs. (5), (6), (10)–(27), $\alpha=0$
Stage 2:	max Z_0	s.t. Eqs. (5)–(7), (10), (15)–(27), (30)–(34), $\alpha=0$
Stage 3:	min Z_1	s.t. Eqs. (5), (6), (10)–(29), (35), $\alpha=1$

As shown in Table 1, a three-stage solving algorithm is developed based on vehicle classes and rescue priorities to solve the above-listed difficulties. Stage 1 does not consider rescue contraflow and obtains the optimal rescue traffic objective function value and the upper limit value of the critical rescue traffic volume for reversing the normal road traffic direction. Stage 2 involves solving the rescue traffic optimization problem of considering rescue contraflow, and obtaining the optimal rescue traffic route during evacuation. Stage 3 involves solving the evacuation traffic optimization problem with reserving the rescue traffic route.

Fig. 1 shows the implementation process of the proposed optimization formulation and three-stage solving algorithm for obtaining an evacuation and rescue traffic optimization result dataset.

Step 1 Input the given evacuation and rescue traffic demands E and R, and call the Cplex solver via the self-developed MATLAB program. Solve the objective function max Z_0 subjected to constraints Eqs. (5) and (6), (10)–(27) to obtain the lower boundary Z_0* of the optimal rescue traffic objective function value. Here, lane contraflows are not adopted in rescue.
Step 2 Take λ as the input parameter and give its initial value λ=0. Lane contraflow is adopted by rescue vehicles via constraints Eqs. (7), (30)–(33).
Step 3 Call Cplex solver and solve the objective function max Z_0 subjected to constraints Eqs. (5)–(7), (10), (15)–(27), (30)–(34) ($\alpha = 0$); then, the optimal objective function value Z_0 is obtained under a given parameter λ.
Step 4 Judging whether Z_0 is more than Z_0*. If $Z_0 \geq Z_0$*, we perform **Step 5**. Otherwise, we return to **Step 1** and input new evacuation and rescue traffic demand E and R.
Step 5 The optimal rescue traffic route obtained by **Step 3** during the evacuation is reserved, e.g., $L_R, t_i^+, t_i^-, t^{n,+}_{(i,j)}, t^{n,-}_{(i,j)} (n \in N, i,j \in L_R, j \in \Gamma_i^+)$. The Cplex solver is used to solve the objective function min Z_1 subjected to constraints Eqs. (5),(6), (10)–(29), (35) ($\alpha = 1$), and the optimal evacuation traffic route is obtained considering rescue traffic contraflow and rescue traffic priority.
Step 6 Update the rescue contraflow control parameter λ by using the formulation λ=λ+1 to control the rescue contraflow on the road network, and return to **Step 3.**

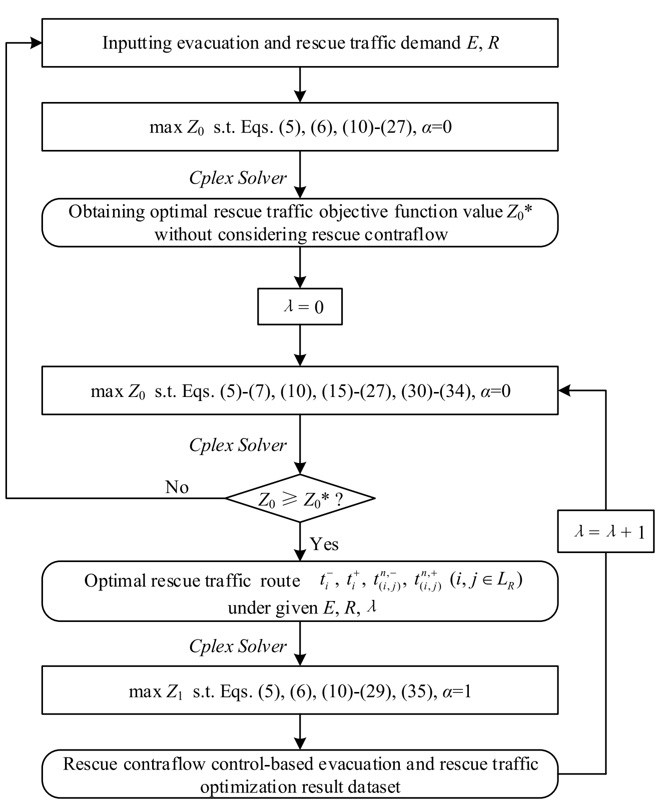


Fig. 1. Flowchart of obtaining the rescue contraflow control-based evacuation and rescue traffic optimization result dataset.

Finally, a large-scale evacuation and rescue traffic optimization result dataset is obtained considering the different traffic demands and rescue traffic contraflow plans.

4. Numerical results

According to the flowchart in Fig. 1, different rescue contraflow control plans are tested on the real Nguyen–Dupuis community road network [22] under various evacuation and rescue traffic demands. The impacts of rescue contraflows on evacuation and rescue traffic and how to control rescue contraflow are investigated based on data-driven statistical analysis to realize collaborative evacuation and rescue traffic operations.

4.1 The Nguyen–Dupuis road network

Fig. 2 shows the topological structure of the Nguyen–Dupuis road network, one of the most representative dynamic emergency traffic assignment networks in existing studies [8,16,23]. In Fig. 2, there are 15 nodes, including 13 intersection nodes, 1 disaster position, 1 safe area, and 46 roads, including 38 actual roads and 8 virtual roads. Here, the traffic flow parameters and the road network structure data are adopted as follows:

(1) The lengths of roads 1~2, 9~16, 27~32, and 37~38 are 200 m; the lengths of roads 3~6, 19~26, and 35~36 are 400 m; the lengths of roads 17~18 and 33~34 are 500 m; the lengths of roads 7~8 are 900 m, and the lengths of 8 virtual roads connecting the disaster position and the outside safe street area are not considered.
(2) All the actual roads have 2 lanes, the road capacity in every lane is 2160 vehicles/hour, the free-flow traffic speed is 72 km/hour, the propagation speed of traffic congestion shockwaves is 18 km/hour, and the jam traffic density is 150 vehicles on every lane every kilometer. Road capacity on virtual roads is not limited.
(3) In response to a local sudden disaster, affected people can leave the disaster position, enter the road network via intersections 1 and 4, and then enter the outside safe street area via intersections 2 and 3; additionally, rescue vehicles can enter the road network via intersections 2 and 3 from the outside safe street area and then arrive at the disaster position via intersections 1 and 4.
(4) The time window for updating the evacuation and rescue traffic state is 15 min, and any discrete period is 10 s. To obtain reliable results and conclusions based on the proposed model and algorithm, we tested 120 different evacuation and rescue traffic demand scenarios involving $R = 30, 60, \ldots, 180$ and $E = 30, 60, \ldots, 600$ under different rescue contraflow control parameters to obtain a large-scale evacuation and rescue traffic optimization result dataset.

4.2 Rescue contraflow control plans

Fig. 3 shows the maximum value of the rescue contraflow control parameter λ under the different rescue traffic demands. Here, when λ reaches these maximum values, there are no contraflow roads for rescue vehicles, all roads are in the normal traffic direction, and the optimal rescue traffic objective function value is the same as the optimal value of not considering rescue contraflow under the same rescue traffic demand. As shown in Fig. 3, with the increase in rescue traffic demand, the

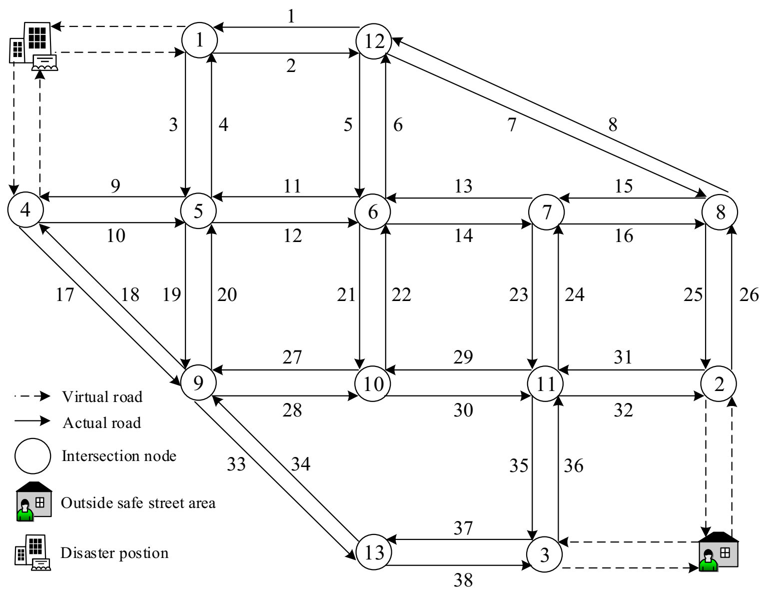


Fig. 2. Nguyen–Dupuis road network.

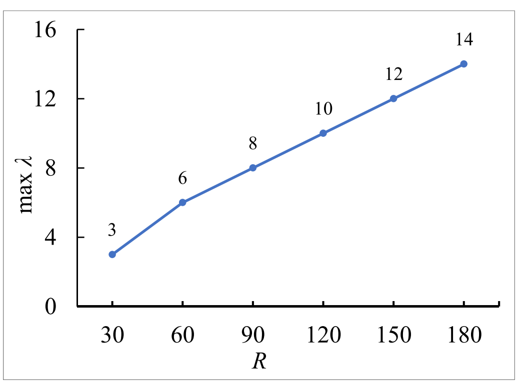


Fig. 3. Maximum rescue contraflow control parameter λ.

maximum value of λ increases, which indicates that the higher the rescue traffic demand is, the greater the critical rescue traffic volume for not reversing the normal road traffic direction under rescue contraflow control. That is, lane contraflow can guarantee quicker arrival of rescue vehicles in the disaster position; compared with low rescue traffic demand, rescue contraflow control can play a more important role in improving rescue traffic operation performance for high rescue traffic demand.

Table 2 presents the number of contraflow roads allocated to rescue vehicles under different rescue contraflow control parameter values and various rescue traffic demands. First, when the rescue contraflow control parameter λ reaches its maximum value, as shown in Fig. 3, no roads are reversed, and rescue vehicles travel in the normal road traffic direction. Second, if rescue vehicles use contraflow roads to improve rescue traffic operation performance, the minimum number of contraflow roads is 4, e.g., $R = 30$, λ=2; $R = 60$, λ=5; $R = 90$, λ=7; $R = 120$, λ=9; $R = 150$, λ=11; and $R = 180$, λ=13. Third, if rescue vehicles use contraflow roads under high rescue traffic demands (e.g., $R = 90$, 120, 150, 180), the minimum number of contraflow roads is 9, except for 4, e.g., $R = 90$, λ=3; $R = 120$, λ=5; $R = 150$, λ=6; and $R = 180$, λ=7. These results show the following: (1) if the critical rescue traffic volume for reversing normal road traffic direction is very high, roads will not be reversed, and lane contraflows will not work to improve rescue traffic operation performance; (2) if roads are reversed for rescue vehicles under lane contraflow, there will be the same minimum number of contraflow roads among different rescue traffic demands. (3) If some roads are reversed for rescue vehicles under high rescue traffic demand (e.g., rescue traffic demand is 90, 120, 150, 180), the same subminimum number of contraflow roads also exists with decreasing critical rescue traffic volume.

Here, we consider the situation in which the normal road traffic direction was reversed for the passage of rescue vehicles. As shown in Table 2, we determined that the minimum number of contraflow roads is 4 among the different rescue traffic demands, and the subminimum number of contraflow roads is 9 among the high rescue traffic demands on the classical Nguyen–Dupuis road network. Therefore, Table 3 further presents the serial numbers of these rescue contraflow roads.

As shown in Table 3, first, when the rescue contraflow control parameter λ reaches its maximum value in the situation that the normal road traffic direction is reversed for rescue vehicles, there are four identical contraflow roads among the different rescue traffic demands,

Table 2
The number of contraflow roads allocated to rescue vehicles.

R	λ														
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
30	4	4	4	0	–	–	–	–	–	–	–	–	–	–	–
60	8	7	7	4	4	4	0	–	–	–	–	–	–	–	–
90	9	12	11	9	4	4	4	4	0	–	–	–	–	–	–
120	13	11	11	10	9	9	4	4	4	4	0	–	–	–	–
150	12	12	12	12	10	9	9	4	4	4	4	4	0	–	–
180	15	12	12	12	11	10	9	9	4	4	4	4	4	4	0

Note: "–" indicates that λ is bigger than its maximum value under the corresponding rescue traffic demand, and lane contraflow is not considered for rescue vehicles.

Table 3
The serial number of rescue contraflow roads.

R	λ	Contraflow roads	λ	Contraflow roads
30	–	–	2	17, 28, 30, 32
60	2	–	5	17, 28, 30, 32
90	3	10, 12, 14, 16, 17, 25, 28, 30, 32	7	17, 28, 30, 32
120	5	10, 12, 14, 16, 17, 25, 28, 30, 32	9	17, 28, 30, 32
150	6	10, 12, 14, 16, 17, 25, 28, 30, 32	11	17, 28, 30, 32
180	7	10, 12, 14, 16, 17, 25, 28, 30, 32	13	17, 28, 30, 32

for example, when the rescue traffic demands are 90 and 120, the four contraflow roads are 17, 28, 30, and 32 under two different maximum rescue contraflow control parameters, λ=7 and λ=9. Second, when the rescue contraflow control parameter λ reaches its maximum value in the situation that the number of contraflow roads is 9, the 9 contraflow roads are the same among the different high rescue traffic demands, for instance, when the rescue traffic demands are 90 and 120, the rescue contraflow control parameter λ is equal to 3 and 5, and the corresponding serial numbers of the 9 contraflow roads are 10, 12, 14, 16, 17, 25, 28, 30, and 32. These results show that if roads are reversed for rescue vehicles under rescue contraflow control, the same contraflow roads may exist under the highest critical rescue traffic volume for reversing normal road traffic directions among the different rescue traffic demands. Moreover, under high rescue traffic demand (e.g., R = 90, 120, 150, 180), if roads are also reversed for rescue vehicles under rescue contraflow control, the same contraflow roads also exist under the second highest critical rescue traffic volume for reversing the normal road traffic direction.

In addition, on the Nguyen–Dupuis road network, contraflow roads 17, 28, 30, and 32 are the shortest routes from the outside safe street area to the disaster position, and contraflow roads 10, 12, 14, 16, and 25 are the sub shortest routes. As shown in Table 3, if the road traffic direction needs to be reversed for rescue vehicles under a very high critical rescue traffic volume of reversing normal road traffic direction, the roads constituting the shortest route from the outside safe street area to the disaster position can be used as rescue contraflow roads; if the road traffic direction needs to be reversed for rescue vehicles under a low critical rescue traffic volume of reversing the normal road traffic direction, the roads constituting the subshortest route from the outside safe street area to the disaster position can also be used as rescue contraflow roads.

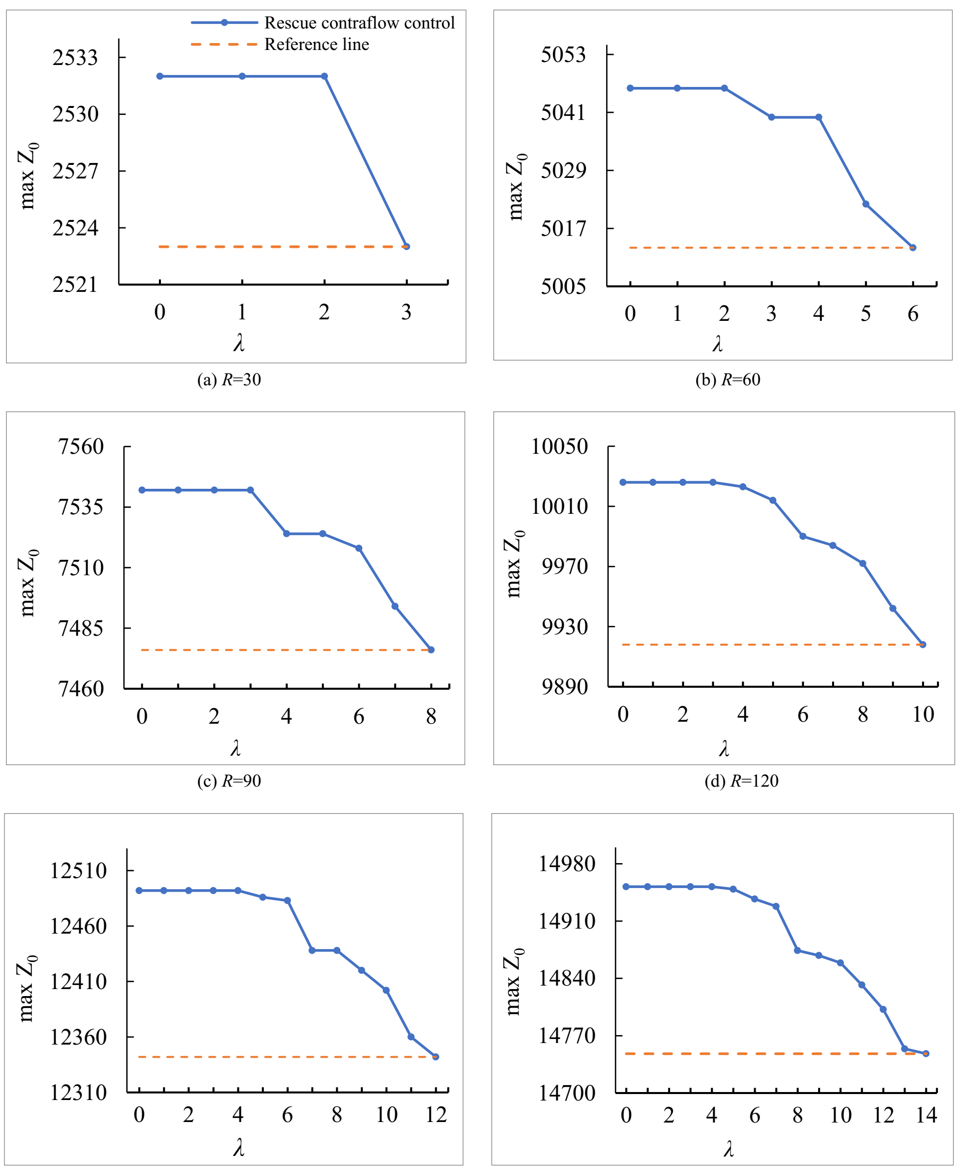


Fig. 4. Optimal rescue traffic objective function values with different rescue traffic control parameter values.

4.3 Rescue traffic operation performance

This section adopts the optimal rescue traffic objective function values presented in Fig. 4 to assess the rescue traffic operation performance. In Fig. 4, “Reference line” represents the optimal rescue traffic objective function value Z_0* without considering rescue contraflow, and “Rescue contraflow control” represents the optimal rescue traffic objective function value Z_0 when considering different rescue contraflow control plans. Here, the crossover point between the “Reference line” and “Rescue contraflow control” states that the optimal rescue traffic objective function value of considering rescue contraflow control is equal to the optimal value of not taking rescue contraflow into account, which shows that although lane contraflow is integrated into rescue traffic optimization, the traffic direction on all roads is not reversed because meeting the very high critical rescue traffic volume of reversing normal road traffic direction does not improve rescue traffic operation performance.

As shown in Fig. 4, if lane contraflow is adopted to improve the traffic supply in the rescue direction, the optimal rescue traffic objective function values overall decrease with increasing critical rescue traffic volume of reversing the normal road traffic direction under different rescue traffic demands. Thus, if higher critical rescue traffic volumes are required to reverse the normal road traffic direction, the improvement in the rescue traffic operation performance of lane contraflow drops. Second, with the increase in the rescue contraflow control parameter λ, the change in the optimal rescue traffic objective function value can be divided into two trends: when the rescue contraflow control parameter λ is small, the optimal rescue traffic objective function value does not change, and the higher the rescue traffic demand is, the greater the scope of the rescue contraflow control parameter λ to maintain the optimal rescue traffic objective function value unchanged; when the rescue contraflow control parameter λ is large, the optimal rescue traffic objective function value begins to decrease almost exponentially. These results show that rescue traffic operation performance is more sensitive to high critical rescue traffic volume of reversing normal road traffic direction and low rescue traffic demands.

4.4 Evacuation traffic operation performance

This section adopts the optimal evacuation traffic objective function

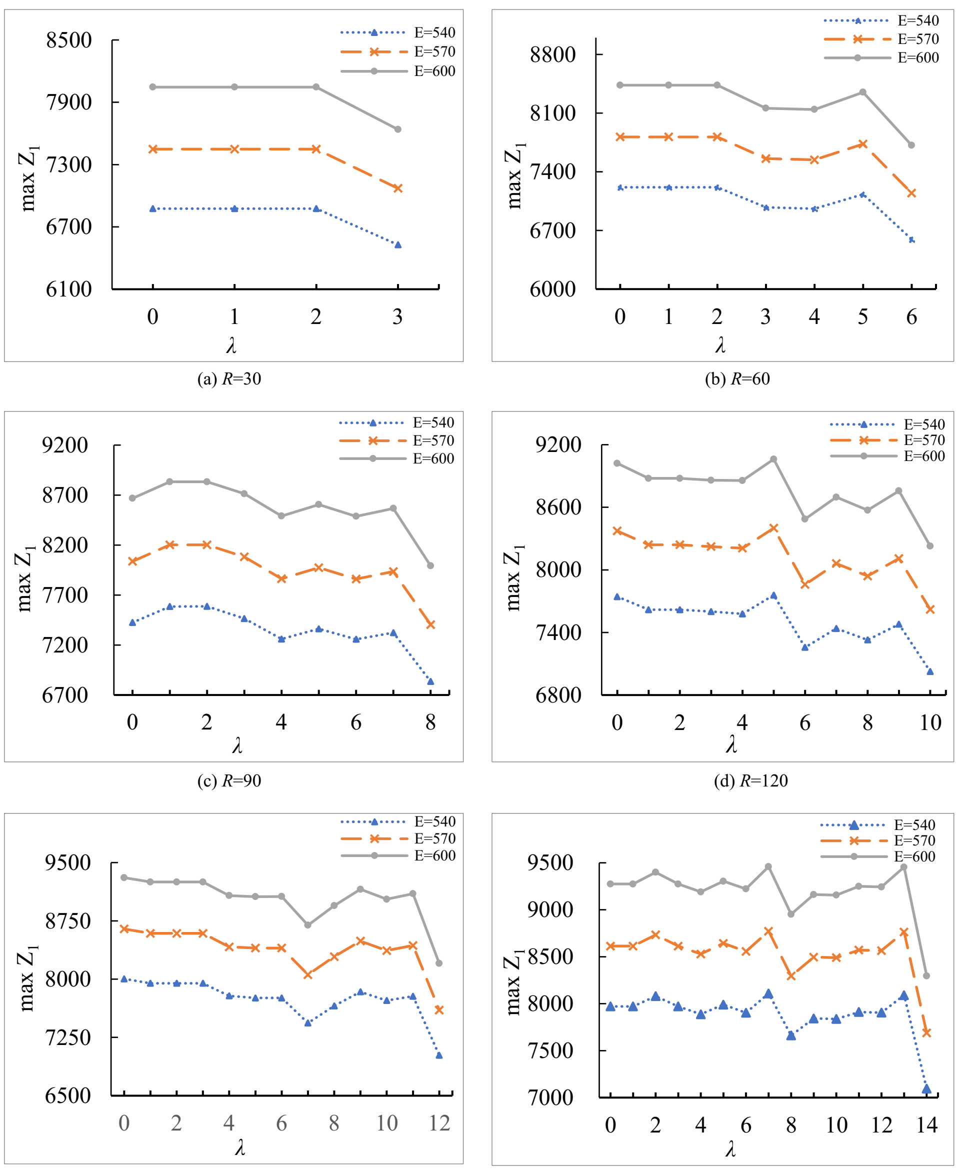


Fig. 5. Optimal evacuation traffic objective function values under different rescue traffic demands.

value to assess the evacuation traffic operation performance. Considering that the change in the optimal evacuation traffic objective function value shows almost the same trend among different evacuation traffic demands under the same rescue traffic demand, Fig. 5 takes three large evacuation traffic demands, $E = 540$, 570, and 600, as examples to analyze how the impact of rescue contraflow on rescue traffic affects evacuation traffic operation performance and investigate the solutions for coordinating evacuation and rescue traffic operation under lane contraflow. Here, another reason for choosing $E = 540$, 570, and 600 is that the high evacuation traffic demand is closer to the supply capacity of the road network and can reflect more complex traffic situations.

As shown in Fig. 5, with the change in the rescue contraflow control parameter λ, the optimal evacuation traffic objective function value shows a decreasing trend overall. For instance, in Fig. 5(d), when the rescue traffic demand is 120, the optimal evacuation traffic objective function value drops to 8228 from 9021 under $E = 600$ as the rescue contraflow control parameter λ increases to 6 from 0. These results, presented in Figs. 4 and 5, show that, in general, increasing the critical rescue traffic volume for controlling rescue contraflow to restrict the occupancy of more road capacity by rescue vehicles reduces rescue traffic operation performance and improves evacuation traffic operation performance.

Differing from the change in the optimal rescue traffic objective function value presented in Fig. 4, the overall trend of the optimal evacuation traffic objective function value is the fluctuating decrease rather than the continuous decrease with the change of the rescue contraflow control parameter λ. Moreover, the fluctuation becomes more obvious with the increase of rescue traffic demand. Second, taking Figs. 4(d) and 5(d) as examples, when the rescue traffic demand is 120 and the evacuation traffic demand is 600, the optimal rescue traffic objective function value is 10,026 and does not decrease with an increase of the rescue contraflow control parameter λ to 3 from 0, but the corresponding optimal evacuation traffic objective function value drops to 8859 from 9021, which shows that increasing the critical rescue traffic volume of reversing the normal traffic direction can reduce the impact of rescue traffic on evacuation traffic and improve evacuation performance, In contrast, rescue performance does not decrease. Therefore, controlling rescue contraflow can coordinate evacuation and rescue traffic operations. Third, taking Figs. 4(c) and 5(c) as examples, when the rescue traffic demand is 90, and the evacuation traffic demand is 600, although the optimal rescue traffic objective function value does not change with the adjustment of the rescue contraflow control parameter λ to 3 from 0, the optimal evacuation traffic objective function value increases, which shows that if critical rescue traffic volumes for controlling rescue contraflows are adopted unreasonably, evacuation traffic operation performance may also be reduced.

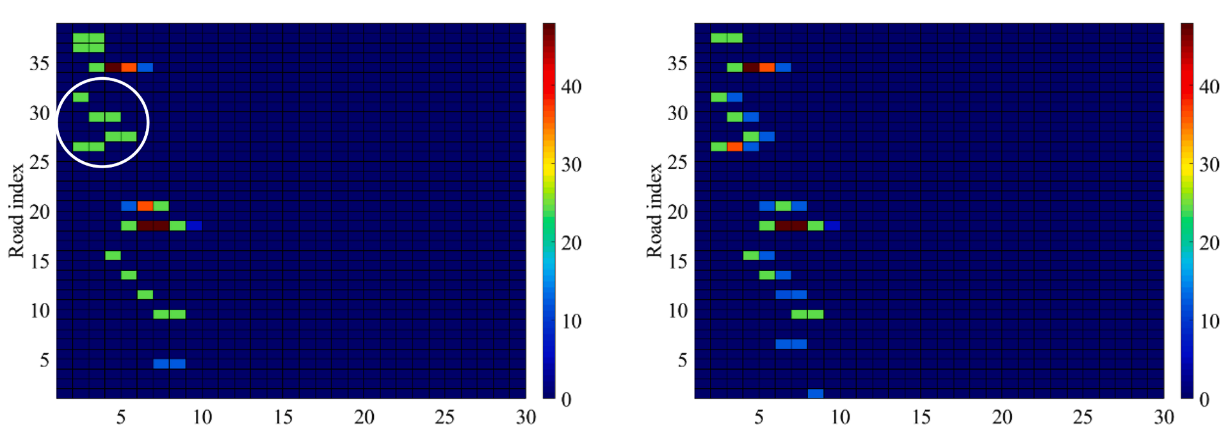
4.5 Evacuation and rescue traffic routes

In this section, we take $E = 600$ and $R = 120$ as examples to analyze the optimal evacuation and rescue traffic routes between λ=0 and λ=3. In Fig. 6(a) and (b), the number of rescue vehicles on the road network during different simulation periods is presented with rescue contraflow control parameters λ=0 and λ=3. In Fig. 7(a) and (b), the number of evacuation vehicles on the road network during different simulation periods is also presented, in which the rescue contraflow control parameters λ=0 and λ=3 are adopted, and the optimal rescue traffic route is reserved during the evacuation.

As shown in Fig. 6(a) and (b), the spatiotemporal distributions of rescue vehicles are significantly different between λ=0 and λ=3 under the same rescue traffic demand, for example, from the beginning of the 2nd simulation period to the end of the 6th simulation period, the numbers of rescue vehicles are different between λ=0 and λ=3 on roads 26, 27, 29, and 31. As shown in Fig. 7(a) and (b), the spatiotemporal distributions of evacuation vehicles also differ on the same roads as the distributions of rescue vehicles change between λ=0 and λ=3; for example, on road 7 connecting intersection 12 and intersection 8, the number of evacuation vehicles significantly differs from the beginning of the 5th simulation period to the end of the 11th simulation period. These results reveal that the critical rescue traffic volume for reversing the normal road traffic direction changes the dynamic distribution of rescue vehicles in the road network, and these changes further affect the dynamic distribution of evacuation vehicles.

Considering the impact of different rescue contraflow control parameters λ on the spatiotemporal distribution of evacuation vehicles and rescue vehicles, the corresponding optimal evacuation and rescue traffic routes are presented in Fig. 8. The red dashed arrow represents the rescue traffic direction from the outside safe street area to the disaster position, and the green solid arrow represents the evacuation traffic direction from the disaster position to the outside safe street area. In addition, there are three numbers near the red dashed arrow and the green solid arrows. The first number represents the simulation period during which the vehicles begin to enter the roads, the second number represents the simulation period during which all vehicles leave the roads, and the third number represents the cumulative number of vehicles entering the roads.

Comparing Fig. 8(a) with Fig. 8(b), the number of evacuation vehicles and rescue vehicles choosing the same traffic route is different with the change in the rescue contraflow control parameter λ between 0 and 3. For example, when the rescue contraflow control parameter λ is set to 0, the number of rescue vehicles entering the route connecting intersections 2, 8, 7, and 6 is 24, and the number of rescue vehicles entering the road connecting intersections 6 and 12 is 0. However, when



(a) R=120, λ=0
(b) R=120, λ=3

Fig. 6. Time-space distribution of 120 rescue vehicles with rescue contraflow control parameters λ=0 and λ=3.

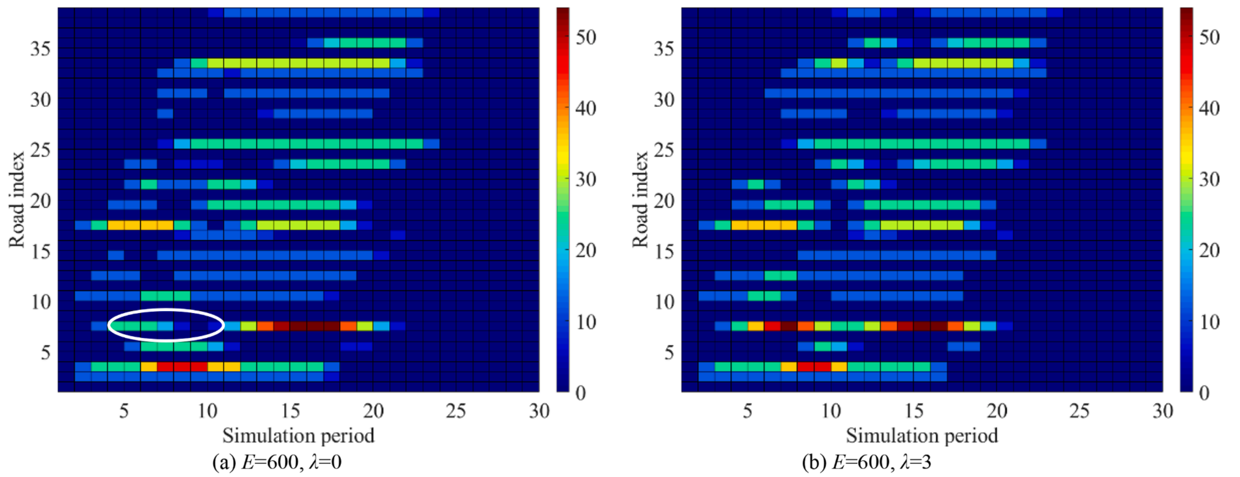


Fig. 7. Time-space distribution of 600 evacuation vehicles with rescue traffic demand $R = 120$ and rescue contraflow control parameters λ=0 and λ=3.

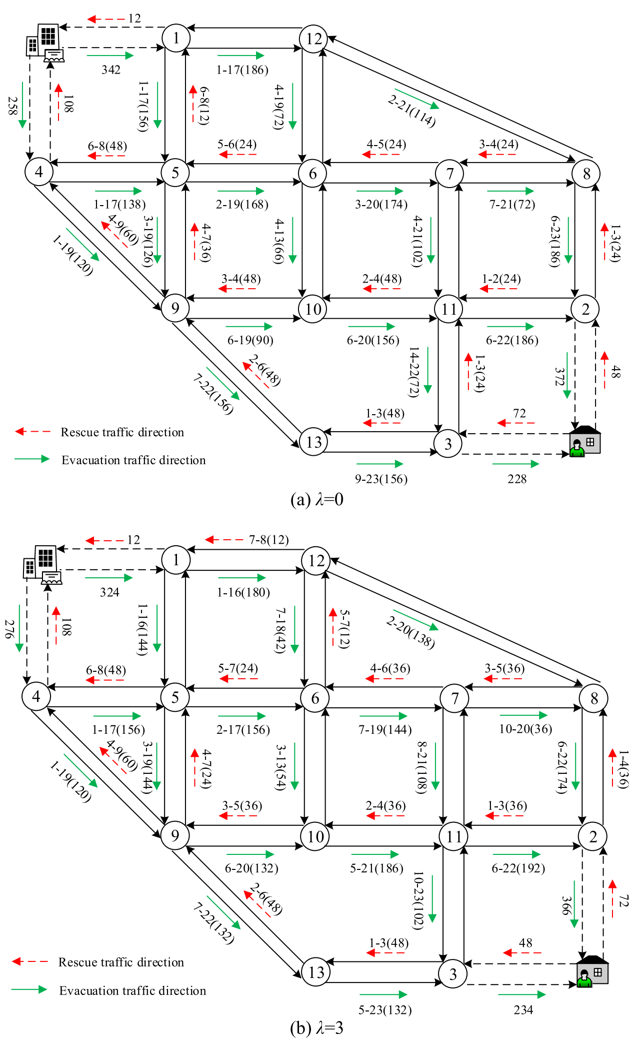


Fig. 8. Optimal evacuation and rescue traffic routes with $R = 120$ and $E = 600$ for λ=0 and λ=3.

the rescue contraflow control parameter λ is set to 3, the number of rescue vehicles entering the route connecting intersections 2, 8, 7, and 6 is 36, and the number of rescue vehicles entering the road connecting intersections 6 and 12 is 12. These results show that (1) the optimal rescue traffic route is not unique under the same rescue traffic operation performance during the evacuation, and different optimal rescue traffic routes can be obtained by adjusting the critical rescue traffic volume of reversing the normal road traffic direction; (2) the impact of controlling rescue contraflow on the optimal rescue traffic route affects the optimal evacuation traffic route.

5. Conclusions

This paper considered rescue contraflow increases the rescue traffic supply but reduces the traffic supply in the evacuation direction on a shared road network; furthermore, multiobjective evacuation and rescue traffic optimization formulation considering rescue contraflow control was proposed, and a three-stage solving algorithm was designed. The corresponding implementation process for obtaining large-scale evacuation and rescue traffic optimization results were presented to obtain reliable results and conclusions. Based on the proposed MMILP formulation and algorithm, the impact of rescue contraflow on evacuation and rescue traffic operation was studied by adjusting the critical rescue traffic volume of reversing the normal road traffic direction from the perspective of data-driven statistical analysis. Some novel findings were captured on the classical Nguyen–Dupuis road network:

(1) As the rescue traffic demand increases, the critical rescue traffic volume needed to avoid reversing the normal road traffic direction increases under lane contraflow. If some roads were used as contraflow roads for rescue vehicles, there would be the same minimum number of rescue contraflow roads among the different rescue traffic demands, and the minimum value would be equal to the number of roads included on the shortest route from the outside safe street area to the disaster position. In addition, the same subminimum number of rescue contraflow roads also exist among high rescue traffic demands. Taking our test on the Nguyen–Dupuis road network as an example, the same minimum number is 4 under rescue traffic demands of 30, 60, …, 180, and the same subminimum number is 9 under high rescue traffic demands of 90, 120, 150, 180.
(2) Under the highest critical rescue traffic volume of reversing the normal road traffic direction, the rescue contraflow roads are the same among the different rescue traffic demands. Taking our test on the Nguyen–Dupuis road network as an example, these same rescue contraflow roads are 17, 28, 30, and 32 and are located on the shortest route from the outside safe street area to the disaster position. That is, the roads constituting the shortest route from the outside to the disaster position can be used as rescue contraflow roads under very high critical rescue traffic volume. In

addition, the roads constituting the sub-shortest route from the outside to the disaster position can also be used as rescue contraflow roads under low critical rescue traffic volume.

(3) As the critical rescue traffic volume increases of reversing the normal road traffic direction, the rescue traffic operation performance decreases overall, and this decrease is more sensitive to high critical rescue traffic volume and low rescue traffic demand. The evacuation traffic operation performance generally has a fluctuation improvement with the increase of the critical rescue traffic volume of reversing the normal road traffic direction. Moreover, the fluctuations are more obvious with increasing rescue traffic demand.
(4) The critical rescue traffic volume of reversing the normal road traffic direction can reduce the impact of rescue traffic on evacuation traffic operation performance when the rescue traffic operation performance is not reduced. However, if the critical rescue traffic volume is not unreasonably adopted, evacuation and rescue traffic operation performance may also be reduced simultaneously.
(5) The optimal rescue traffic route is not unique under the same rescue traffic operation performance during an evacuation, and different optimal rescue traffic routes can be obtained by adjusting the critical rescue traffic volume of reversing the normal road traffic direction; moreover, the impact of controlling rescue contraflow on the optimal rescue traffic routes affects the optimal evacuation traffic route.

Based on this paper, in the future, we will study new algorithms from the perspective of data-driven statistical analysis to solve evacuation traffic optimization problems and rescue traffic optimization problems simultaneously, including multiobjective evacuation and rescue traffic collaborative optimization formulations considering unknown parameters. Then, by combining the evacuation and rescue traffic optimization model and algorithm, we plan to use big data prediction technologies to optimize evacuation and rescue traffic operations based on a large-scale evacuation and rescue traffic optimization result dataset.

Ethics statement

Not applicable because this work does not involve the use of animal or human subjects.

CRediT authorship contribution statement

Zheng Liu: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jialin Liu:** Conceptualization, Validation, Writing – original draft, Writing – review & editing. **Xuecheng Shang:** Conceptualization, Validation, Writing – review & editing. **Xingang Li:** Conceptualization, Funding acquisition, Investigation, Supervision, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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